

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Arava Raja (192365033) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Arava Raja

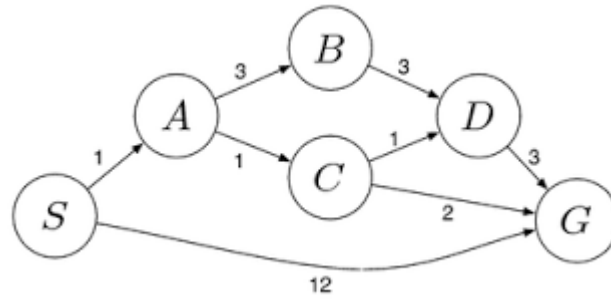
Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1. Water Jug Problem (4-gallon and 3-gallon jugs)

Initial State

- 4-gallon jug = 0 gallons
- 3-gallon jug = 0 gallons
-

Goal

Obtain exactly **2 gallons in the 4-gallon jug**.

Step 4-Gallon Jug 3-Gallon Jug Action

Start	0	0	Initial state
1	0	3	Fill 3-gallon jug
2	3	0	Pour into 4-gallon jug
3	3	3	Fill 3-gallon jug again
4	4	2	Pour into 4-gallon until full
5	0	2	Empty 4-gallon jug
6	2	0	Pour remaining 2 gallons into 4-gallon jug

Goal Achieved

4-gallon jug = 2 gallons

2. Mars Rover Intelligent Agent

(i) Percepts

- Images captured by cameras
- Soil and rock samples
- Temperature
- Atmospheric pressure
- Radiation level
- Terrain information
- Obstacle detection
- GPS/Position information

(ii) Operating Environment

- **Partially Observable** – cannot observe the whole environment.
- **Stochastic** – uncertain weather and terrain.
- **Sequential** – every action affects future actions.
- **Dynamic** – environment changes with time.
- **Continuous** – movement and sensing are continuous.
- **Single Agent** – only one rover performs the task.

(iii) Actions

- Move forward/backward
- Turn left/right
- Capture images
- Collect rock samples
- Drill rocks
- Analyze samples
- Send data to Earth
- Avoid obstacles

(iv) Performance Measures

- Successful navigation
- Number of samples collected
- Accuracy of analysis
- Battery efficiency
- Time taken
- Safety (avoid collisions)
- Reliable communication

(v) Suitable Agent Architecture

Utility-Based Learning Agent

Reason:

- Learns from previous experiences.
- Makes optimal decisions.
- Handles uncertain environments.
- Maximizes scientific discoveries while minimizing energy consumption.

3. 8-Queens Problem Formulation

Initial State

Empty chessboard.

State Space

All possible arrangements of 8 queens.

Operators

- Place one queen in a safe position.
- Move queen to another row if conflict occurs.

Goal Test

No two queens attack each other.

Conditions:

- Same row
- Same column
- Same diagonal

Path Cost

Each queen placement costs 1.

Search Strategy

Backtracking Search (Depth First Search)

Algorithm:

1. Place queen in first column.
2. Check safety.
3. If safe, place next queen.
4. If unsafe, backtrack.
5. Continue until all eight queens are placed.

Example Solution

Q
 Q . . .
 Q
 Q . .
 . . Q
 Q .
 . Q
 . . . Q

No queens attack one another.

4. OLA Cab Booking Problem-Solving Agent

Agent Type
Goal-Based Problem-Solving Agent
Reason

The customer has a goal:

- Reach destination safely
- Choose preferred cab
- Minimize travel time and cost

Pseudocode

Algorithm OLA_Cab_Booking

Start

Input Source

Input Destination

Display Available Cab Types

(Mini, Micro, Sedan, Prime, Shared)

User selects Cab Type

Calculate Distance

Estimate Fare

Display Fare

If User Confirms Booking

 Assign Nearest Driver

 Send Driver Details

 Track Ride

 Complete Payment

Else

Cancel Booking

End If

Stop

5. Uniform Cost Search (UCS) Algorithm

UniformCostSearch (Start, Goal)

1. Create Priority Queue

2. Insert Start node with cost 0

3. While queue is not empty

Remove node with minimum cost

If Goal reached

Return Path

Else

Expand all neighboring nodes

Add cumulative cost

Insert into queue

4. End While

5. Return Failure

Working Principle

- Starts from source node **S**.
- Expands the node with the **lowest cumulative path cost**.
- Continues until destination **G** is reached.
- Guarantees the **least-cost path** when all edge costs are non-negative.

Advantages

- Finds the optimal solution.
- Complete algorithm.
- Works for weighted graphs.
- Guarantees minimum-cost path.

Disadvantages

- Requires more memory.
- Can be slow for very large graphs.

Note: The uploaded document references a weighted graph for Question 5, but the graph itself is not included in the uploaded pages. Therefore, the exact least-cost path from **S** to **G** cannot be computed from the provided file.